

Security+ 4.8 • Log Data

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 4.8 — Log Data (13:41)** · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Log data from various sources (firewall, application, endpoint, OS, network, proxy) must be collected, retained, and protected. Retention policies define how long logs are kept, often driven by legal or audit requirements. Logs must be protected from unauthorized access or modification. Centralizing logs in a SIEM enables correlation and analysis.

High-Yield Core Concepts Tested on Exam

- Collect logs from all sources: firewall, application, endpoint, network, cloud
- Retention: how long logs are kept, often 1 year for compliance
- Protection: logs must be secured against tampering and unauthorized access
- SIEM: centralized log repository for correlation and alerting

Exam Tips & Classic Scenarios

- If asked about a centralized log repository, think SIEM.
- If asked about how long to keep logs, think retention.

Typical Exam Scenario: An organization sends firewall, application, endpoint, and DNS logs to a single platform for analysis. This is log aggregation, typically done in a SIEM.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 4.8 — Log Data)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **SIEM:** Security Information and Event Management: Centralized platform performing real-time ingestion, normalization, correlation, and alerting across multi-source security telemetry.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. A tool that collects and CORRELATES logs from multiple systems to find threats is:

- A)** Firewall
- B)** SIEM
- C)** Proxy
- D)** Antivirus

2. Corporate scenario (Log Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.8), an analyst evaluates the following technical requirement: 'Collect logs from EVERYTHING: firewall+app+endpoint+network+cloud'. What is the best recommendation or classification of applicable control?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: Collect logs from EVERYTHING: firewall+app+endpoint+network+cloud...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

3. Corporate scenario (Log Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.8), an analyst evaluates the following technical requirement: 'Retention by law: 1 year is common; log protected against alteration.' What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators using generic credentials
- D)** Implement the corresponding solution: Retention by law: 1 year is common; log protected against alteration...

4. Corporate scenario (Log Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.8), an analyst evaluates the following technical requirement: '"centralized log repository'=SIEM; 'how long to keep'=retention'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: 'centralized log repository'=SIEM; 'how long to keep'=retention...
- D)** Allow shared access among administrators using generic credentials

5. Corporate scenario (Log Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.8), an analyst evaluates the following technical requirement: 'firewall, application, endpoint, and DNS logs all in the same place'=log aggregation. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'firewall, application, endpoint, and DNS logs all in the same place'=log aggregation...

6. Corporate scenario (Log Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.8), an analyst evaluates the following technical requirement: 'Collect logs from EVERYTHING: firewall+app+endpoint+network+cloud'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: Collect logs from EVERYTHING: firewall+app+endpoint+network+cloud...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

7. Corporate scenario (Log Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.8), an analyst evaluates the following technical requirement: 'Retention by law: 1 year is common; log protected against alteration'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: Retention by law: 1 year is common; log protected against alteration...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

8. Corporate scenario (Log Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.8), an analyst evaluates the following technical requirement: 'centralized log repository'=SIEM; 'how long to keep'=retention. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: 'centralized log repository'=SIEM; 'how long to keep'=retention...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

9. Corporate scenario (Log Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.8), an analyst evaluates the following technical requirement: 'firewall, application, endpoint, and DNS logs all in one place' = log aggregation'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: 'firewall, application, endpoint, and DNS logs all in one place' = log aggregation.
- D)** Allow shared access among administrators using generic credentials

10. Corporate scenario (Log Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.8), an analyst evaluates the following technical requirement: 'Collect logs from EVERYTHING: firewall+app+endpoint+network+cloud'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: Collect logs from EVERYTHING: firewall+app+endpoint+network+cloud...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) B — SIEM.

✓ **Why it's correct:** The correct answer is SIEM (Security Information and Event Management). In the CompTIA Security+ SY0-701 Security Architecture and Operations domain, SIEM is the central platform that aggregates, normalizes, and correlates logs and events from multiple heterogeneous sources (firewalls, servers, endpoints, applications, IDS/IPS, routers, etc.). Correlation is the core of SIEM: it applies rules, signatures, and increasingly behavioral analytics (UEBA) to identify patterns that would not be detected in isolation. For example, a failed login on a Linux server followed by a successful access to a file share and an unusual outbound connection to an external IP can be correlated as possible lateral movement and exfiltration. SIEM also provides dashboards, real-time alerts, log retention for compliance, and support for forensic investigations. It is the tool specifically designed for the function described in the question: collecting and correlating logs from multiple systems to find threats.

✗ **Why the others are wrong:**

- *Firewall:* A firewall is a network security device that filters traffic based on access control rules (IPs, ports, protocols, connection states). It can generate logs of allowed and denied connections, but its primary function is to enforce perimeter or segmentation security policies, not to collect and correlate logs from other systems. A firewall does not aggregate logs from servers, endpoints, or applications; it only records its own events. Therefore, it is incapable of performing the multi-source correlation necessary to detect complex threats that span different layers of the infrastructure.
- *Proxy:* A proxy acts as an intermediary for network requests, typically at the application layer (HTTP/HTTPS), providing access control, caching, content filtering, and anonymization. It generates web traffic logs, but these logs are limited to the scope of the traffic that passes through it. A proxy does not collect logs from operating systems, databases, authentication, or other security devices. Its correlation capability is restricted to its own domain and does not replace a SIEM, which integrates and correlates events from the entire infrastructure.
- *Antivirus:* Antivirus is an endpoint solution that detects and removes malware based on signatures, heuristics, and local behavioral analysis. It generates detection logs on the host itself, but it does not collect or correlate logs from other systems. Its view is limited to the device where it is installed and does not cover the infrastructure as a whole. Although it can send alerts to a SIEM, it does not itself perform the multi-source correlation function described in the question.

2) B — Implement the corresponding solution: Collect logs from EVERYTHING: firewall+app+endpoint+network+cloud....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Log Data): Collect logs from EVERYTHING: firewall+app+endpoint+network+cloud.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

3) D — Implement the corresponding solution: Retention by law: 1 year is common; log protected against alteration....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Log Data): Retention by law: 1 year is common; log protected against alteration.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

4) C — Implement the corresponding solution: 'centralized log repository'=SIEM; 'how long to keep'=retention....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Log Data): 'centralized log repository'=SIEM; 'how long to keep'=retention. A SIEM (Security Information and Event Management) system aggregates and correlates logs from multiple sources, while retention defines the duration logs are stored to meet compliance and forensic requirements.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

5) D — Implement the corresponding solution: 'firewall, application, endpoint, and DNS logs all in the same place'=log aggregation....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Log Data): 'firewall, application, endpoint, and DNS logs all in the same place'=log aggregation.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

6) A — Implement the corresponding solution: Collect logs from EVERYTHING: firewall+app+endpoint+network+cloud....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Log Data): Collect logs from EVERYTHING: firewall+app+endpoint+network+cloud. Comprehensive log collection from all sources is essential for security monitoring, incident detection, and forensic analysis.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

7) A — Implement the corresponding solution: Retention by law: 1 year is common; log protected against alteration....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Log Data): Retention by law: 1 year is common; log protected against alteration. Many regulations require log retention for a specific period (e.g., 1 year) and logs must be protected from tampering to ensure integrity for audits and forensics.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

8) B — Implement the corresponding solution: 'centralized log repository'=SIEM; 'how long to keep'=retention....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Log Data): 'centralized log repository'=SIEM; 'how long to keep'=retention. SIEM (Security Information and Event Management) centralizes log collection and analysis, while retention defines how long logs are stored for compliance and forensic purposes.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual

accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

9) C — Implement the corresponding solution: 'firewall, application, endpoint, and DNS logs all in one place' = log aggregation..

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Log Data): 'firewall, application, endpoint, and DNS logs all in one place' = log aggregation. Log aggregation centralizes logs for efficient analysis and correlation; the appropriate control is to implement the corresponding solution, such as a SIEM system.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

10) B — Implement the corresponding solution: Collect logs from EVERYTHING: firewall+app+endpoint+network+cloud....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Log Data): Collect logs from EVERYTHING: firewall+app+endpoint+network+cloud. Comprehensive log collection from all sources enables effective security monitoring, incident detection, and forensic analysis. Centralized logging (SIEM) is essential for correlation and compliance.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.